Project Design Phase-II
Technology Stack (Architecture & Stack)

Date	6 february 2026
Team ID	LTVIP2026TMIDS86321
Project Name	IntelliSQL: Intelligent SQL Querying with LLMs Using Gemini Pro
Maximum Marks	4 Marks

Technical Architecture:

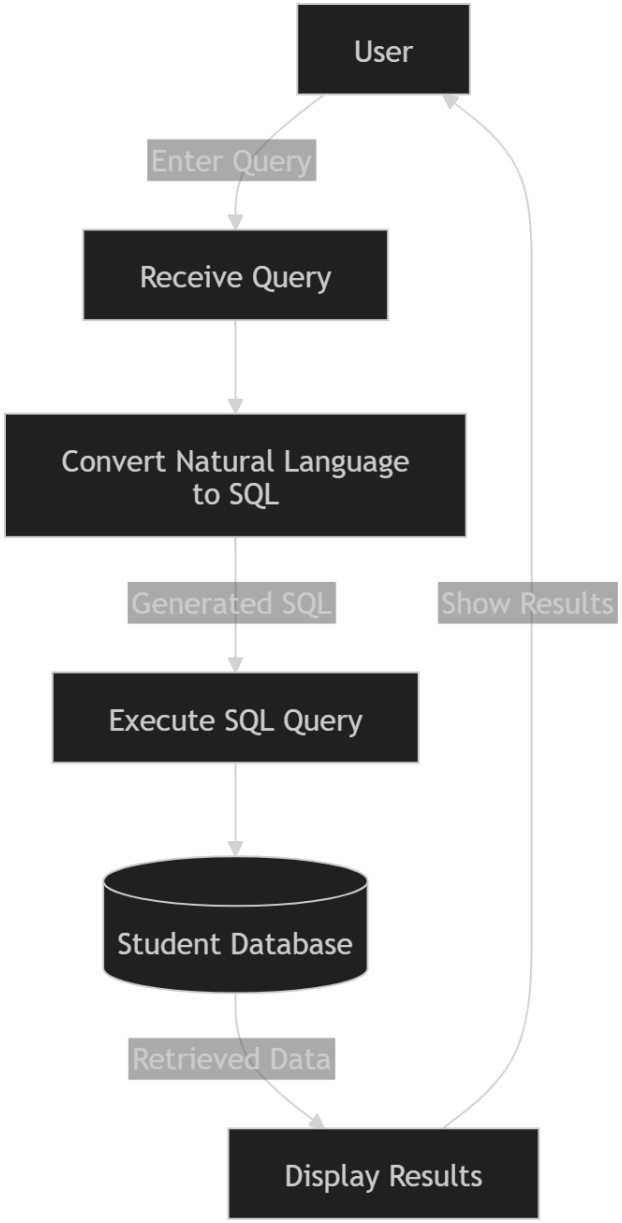
Technical Architecture

IntelliSQL follows a Client–Application Architecture Model consisting of the following layers:

1. **User Interface Layer (Streamlit Web App)**
2. **Application Logic Layer (Python Backend)**
3. **Query Processing Layer (Natural Language to SQL Conversion)**
4. **Database Layer (SQLite Database)**
5. **Deployment Layer (Local System / Streamlit Cloud)**

Example: Order processing during pandemics for offline mode

Reference: https://developer.ibm.com/patterns/ai-powered-backend-system-for-order-processing-during-pandemics/



Guidelines:

1. Fill project details (Date, Team ID, Project Name).
2. Write system architecture layers (UI → Logic → Query Processing → Database → Deployment).
3. Draw the architecture diagram showing data flow.
4. Fill **Table-1** with components and technologies used.
5. Fill **Table-2** with application characteristics like usability, security, performance, and availability.

Table-1: Components & Technologies

S.No	Component	Description	Technology
1	User Interface	Web interface where users enter natural language queries	Streamlit (Python Web Framework)
2	Application Logic-1	Query input processing and validation	Python
3	Application Logic-2	Natural language query conversion to SQL	Python
4	Application Logic-3	Session handling and result display	Streamlit Session State
5	Database Storage	Stores student records and data	SQLite Database
6	Query Execution	Executes generated SQL queries on database	SQLite Engine
7	Data Processing	Handles filtering such as highest, lowest, and average marks	Python
8	Infrastructure	Local development and cloud deployment	Local System / Streamlit Cloud

Table-2: Application Characteristics

S.No	Characteristics	Description	Technology
1	Open-Source Frameworks	Application built using open-source libraries	Python, Streamlit, Pandas
2	Security Implementations	Secure handling of database queries and input validation	Python Input Validation
3	Scalable Architecture	Modular structure separating UI, logic, and database layers	Python-based modular architecture
4	Availability	Web application accessible through browser	Streamlit Cloud / Local Server
5	Performance	Lightweight application with fast query processing	Python + SQLite

References:

https://c4model.com/

https://developer.ibm.com/patterns/online-order-processing-system-during-pandemic/

https://www.ibm.com/cloud/architecture

https://aws.amazon.com/architecture

https://medium.com/the-internal-startup/how-to-draw-useful-technical-architecture-diagrams-2d20c9fda90d